

Lewis NDAMBIRI

Industrial AI & Analytics | Operations & Systems Optimization | Smart Manufacturing

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VALUE PROPOSITION: Industrial engineer and AI/data science specialist focused on converting manufacturing, logistics and operations data into reliable decisions. Combines process-improvement discipline, supply-chain execution, machine-learning delivery and systems thinking to support smart-factory initiatives, KPI design, production visibility, forecasting, automation and AI-enabled operational excellence.

CORE CAPABILITIES

Industrial AI & Analytics: Machine learning, demand forecasting, RAG systems, XGBoost, scikit-learn, local LLM/Ollama, KPI design, data storytelling.

Manufacturing & Operations: Lean manufacturing, 5S, Six Sigma Yellow Belt, OEE, process mapping, production scheduling, layout optimization, Kanban, supply chain analytics, logistics coordination.

Data, Systems & Delivery: Python, SQL, VBA, REST APIs, FastAPI, Streamlit, Power BI, Docker, Linux, PostgreSQL, MQTT, Git/GitHub, CI/testing basics.

Engineering Tools: FlexSim, SolidWorks, AutoCAD, MS Project, SAP, Advanced Excel, C/C++, MPI/OpenMP, VS Code.

PROFESSIONAL EXPERIENCE

Industrial Engineering / AI Automation Intern | BLM Group

Mar 2026 - Present | Levico Terme, Italy

- Designing an agentic AI system for production optimization by integrating data pipelines, KPI logic, process knowledge, and customer-order to sales-delivery workflows to improve components traceability and operational decision support.
- Automating Kanban tracking systems to enhance inventory visibility, replenishment discipline, and operational responsiveness.
- Standardizing processes and data collection frameworks to improve the reuse of production knowledge for analytics and AI applications.
- Benchmarking AI project performance through literature review, process analysis and implementation-roadmap development for industrial AI and R&D initiatives.

Intermodal Sales Support Intern & Cargo Readiness Officer | CMA CGM

Jul 2023 - Aug 2024 | Nairobi, Kenya

- Managed end-to-end shipment-readiness operations across bookings, vendor coordination, stuffing, documentation and bill-of-lading processing.
- Improved cargo space utilization by 15% through advanced load-planning and operational coordination, contributing to cost-saving opportunities.
- Automated Excel/VBA reporting workflows, reducing manual errors and improving cargo-readiness visibility by 20%.
- Maintained 100% compliance with export-clearance, stuffing and loading procedures while coordinating multi-country logistics stakeholders.

Continuous Improvement Engineering Intern | Elis Group

Apr 2022 - Oct 2022 | Aix-les-Bains, France

- Led 5S deployment across linen and consumables storage areas, increasing usable warehouse space by 25%.
- Optimized production layouts and ergonomic workstation flow with cross-functional teams, improving throughput and reducing retrieval time.
- Supported production scheduling, logistics coordination and preventive-maintenance routines to reduce unplanned downtime risk.

Additional Engineering Experience

- Jamizam Enterprises - Project Management & Engineering Intern:** Supervised 40+ workers on road/bridge construction works, supported HSE compliance, topographic surveys, drainage execution and client-contractor coordination.
- Ekiu Group - Industrial Engineering Intern:** Supported P&IDs, feasibility studies, AutoCAD drafting, BoQs and API 650-based compliance support for industrial/oil & gas projects in Mauritius, Sierra Leone and Tanzania.

EDUCATION

EIT Manufacturing Double Degree - M.Sc. Data Science & AI for Competitive Manufacturing | EIT Manufacturing Master School

2024 - 2026 | France & Italy

- Centrale Nantes:** M.Sc. Industrial Engineering - Smart and Connected Enterprise.
- University of Trento:** M.Sc. Computer Science - ICT Innovation.
 - Focus: Industrial AI, data science, smart manufacturing, operations research, simulation, service design, innovation and ICT entrepreneurship.

B.Sc. (Hons) Engineering, Specialization in Industrial Engineering | Centrale Nantes

2018 - 2022 | Nantes, France

- Foundation in industrial engineering, production systems, engineering analysis, project management and applied technical problem-solving.

SELECTED PROJECTS

- VoltWise | AI-Powered Industrial Energy Optimizer** - Built a factory energy analytics concept using Python, scikit-learn (Isolation Forest), Streamlit, and Docker to detect inefficiencies and target 10-20% electricity cost reduction.
- Agentic AI in OSCM** - Developed a research framework and implementation roadmap for autonomous AI agents in use cases such as inventory coordination, sustainable negotiation, digital twins, and supply chain workflows.
- Industrial AI Challenge | Fedrigoni Slitting Process** - Analyzed and modelled the slitting process at Fedrigoni's Arco plant to support predictive production planning and shift decision-making from reactive to predictive management. [Python, Pandas, NumPy, XGBoost, CRISP-DM]
- Offline CNC RAG Assistant | Local AI for Machine Documentation** - Built an offline retrieval-augmented generation assistant for CNC machine operation and maintenance documentation, focused on safe, source-grounded answers rather than generic chatbot behavior. [\[link_1\]](#)
- FactoryPulse | Industrial Production Monitoring & Control** - Built a full-stack smart-factory monitoring system that transforms simulated machine telemetry into a browser HMI/dashboard with live status, OEE analytics, alarms, downtime reasons, reports and audit logging. [\[link_2\]](#)

CERTIFICATIONS, LANGUAGES & INTERESTS

Certifications: Six Sigma Yellow Belt; Advanced Manufacturing; Python for Data Science & AI (IBM); Lean 4.0.

Languages: English (C2), Swahili (C2), French (B2), Italian (A2).

Interests: Industrial AI, emerging technologies, continuous learning, travel and cultural exploration, fitness and sports.

